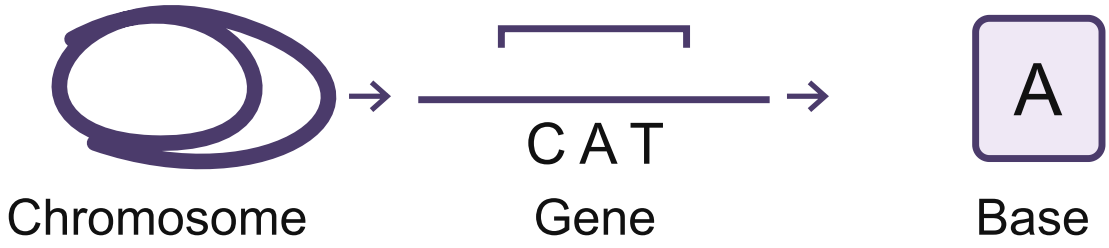
# Reference A zoom drawing



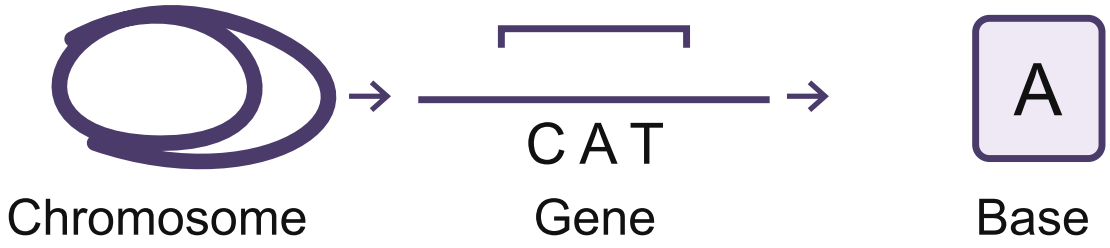

A coiled chromosome, a bracketed gene and the base A.

Reference only. The drawings are simplified and do not show scale.

| Level | What the picture shows |
|---|---|
| Chromosome | A long DNA molecule packaged with proteins. |
| Gene | A bracketed section of the DNA. |
| Base | One unit in the DNA sequence, shown here as A. |

Made by Matt

# Reference B explain the links




The three levels run from chromosome to gene to base.

Reference only.

| Link | Explanation |
|---|---|
| Chromosome to gene | A chromosome carries a long DNA molecule. One region of that DNA can be a gene. |
| Gene to base | The gene has a sequence of bases. A represents one base. |
| Model limit | The drawing leaves out packaging proteins and most bases. The sizes are simplified. |

The genome is the complete DNA set. It includes DNA from all chromosomes.

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# Arrival and exit Aspire

Use the picture on Reference A for question 1 and the model table for questions 2–4.

1. Which drawing is labelled Gene?

2. Why does the gene drawing show less DNA than the chromosome?

3. Why is A shown in the final drawing?

4. What does the zoom leave out?

## Exit ticket

1. Correct “A gene is the whole DNA package”.

2. Name one limit of the zoom model.

**Pupil voice**

What helped you most today, or what would you change next time?

Point, speak, sign, draw, write or choose. You may pass.

verbal · written · drawn · signed · symbol / choice-selected · demonstrated / re-attempt · adult-scribed. You may pass.

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# You do Aspire

Use your book or this sheet. Use Reference A and B. One response per task.

**1. Explain what is lost if a model labels an entire chromosome as a gene.**
**Staff I ☐ S ☐ C ☐**

---

**2. Explain how the base sequence relates to a protein instruction in a protein-coding gene.**
**Staff I ☐ S ☐ C ☐**

---

**3. State one limit of the zoom model and make a justified improvement.**
**Staff I ☐ S ☐ C ☐**

---

# Feedback and response · Aspire

## Feedback codes

| Code | Meaning |
|---|---|
| VF | Verbal feedback |
| WS | Worked with support |
| I | Independent |
| NS + … | Next step |
| E | Evidence captured |
| R | Pupil responded to feedback |
| // | Self-edit point |
| ? | Read this back to me |

## My response

Record one response to feedback in your usual mode. You may pass.

Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

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